

Summary

Second-year Computer Engineering student at TIET with hands-on experience in full-stack development, computer vision, and interactive systems. Built and deployed production-ready platforms used by student teams on campus. Interned at an AI startup building real-time ML pipelines. Looking to solve real problems at the intersection of software and intelligence.

Education

Thapar Institute of Engineering and Technology <i>B.Tech in Computer Engineering — CGPA: 8.61/10</i>	2024 – 2028 (Expected) <i>Patiala, India</i>
Modern Senior Secondary School <i>Class XII – PCM with Computers — Percentage: 92.6%</i>	2022 – 2024 <i>Patiala, India</i>

Experience

Auraflo (Student Startup, TIET Ventures Lab) <i>AI/ML Intern</i>	Mar 2025 – Sep 2025 <i>Remote</i>
- Engineered a real-time pose detection pipeline processing live streams at 30-45 FPS using TensorFlow Lite and OpenCV. - Developed state-transition logic with confidence-based filtering for accurate movement quality evaluation, achieving 95% accuracy on internal live-test trials.	

Projects

AtomQuest — Goal Compliance Portal <i>Next.js 14, React, Supabase, Vanilla CSS (GitHub Live)</i>	May 2026
- Engineered a serverless-safe memory cache within Next.js 14, reducing export streaming overhead by 60% on Vercel. - Developed a timeline-driven compliance escalation engine and an E2E client-encrypted chat, with encryption/decryption handled client-side (no server-side decryption).	
Spatial Console — Gesture-Controlled Structural Simulation <i>React, Three.js, MediaPipe (GitHub)</i>	Feb. 2026
- Integrated gesture-driven interactions at 45–60 FPS into a real-time browser-based 3D system using React, Three.js, and MediaPipe. - Implemented a graph-based structural validation engine using BFS to detect instability and simulate collapse.	
Code Vault — Live Coding Heist Game <i>React, Node.js, MongoDB, Tailwind CSS (GitHub Live)</i>	Jan. 2026
- Deployed a full-stack live coding competition platform handling 30+ concurrent teams with 2-second auto-refresh, 3-second leaderboard polling, and warnings for tab switching to preserve contest integrity. - Built drag-and-drop challenge delivery and administrative tools for contest setup, question distribution, and result management.	

Technical Skills

Languages: C++, Python, JavaScript, HTML, CSS, MATLAB, R
Frameworks/Libraries: React.js, Node.js, FastAPI, MediaPipe, dnd-kit, TensorFlow, OpenCV, Pandas, NumPy, scikit-learn
Cloud / Databases: MongoDB, Firebase, SQL, Netlify, Render, Supabase, Vercel
Developer Tools: Git, Docker, Linux, Postman
Coursework: Data Structure and Algorithm Development, Operating Systems, Relational Database Concepts, Object Oriented Programming, Artificial Intelligence, Computer Networks

Positions of Responsibility

LEAD Society (TIET) <i>Technical Secretary</i>	Oct 2024 – Present <i>Patiala, India</i>
- Built the official LEAD Society website: lead2025.netlify.app using HTML, CSS, and JavaScript; maintained the site and deployment. - Mentored 100+ members in web development and Git workflows; spearheaded technical events for a 500+ student community.	
Modern Senior Secondary School <i>Vice Art & Cultural Captain</i>	2022 – 2023 <i>Patiala, India</i>
Organized 4 school-wide events and anchored 3 ceremonial programs, reaching a cumulative audience of 2,000+ students and faculty.	

Achievements

- Solved **500+** DSA problems on LeetCode, GFG, and CodeChef, with focus on medium-hard problems across trees, graphs, DP, and sliding window.
- Merit III** Scholarship awarded for scoring 95.4 percentile in JEE Mains.
- District Gold Medalist & School Chess Champion:** Recognized for excellence in competitive strategy and performance.